

Conditional probabilities

Def. A and B are events.

Assume that $P(B) \neq 0$. The conditional prob. of A given

B is given by

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

event of
interest

information given

Ex. Roll 2 dice, one is in green, and the other is in red.

Let $A = \{ \text{red die} \geq 4 \}$ and

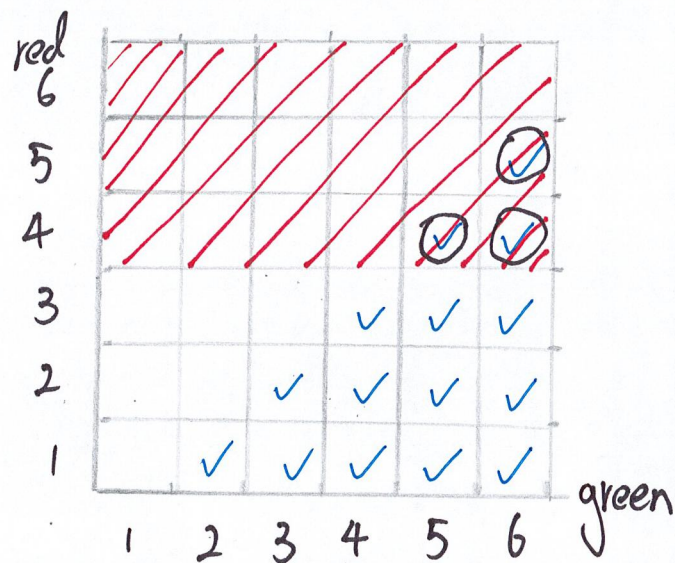
$B = \{ \text{green die} > \text{red die} \}$.

Find $P(A | B)$.

$$P(A \cap B) = \frac{3}{36}$$

$$P(B) = \frac{15}{36}$$

$$P(A|B) = \frac{\cancel{3}/\cancel{36}}{\cancel{15}/\cancel{36}} = \frac{3}{15}$$



$$\boxed{\checkmark} = B$$

$$\boxed{\text{red lines}} = A$$

$$\boxed{\checkmark \text{ in red line}} = A \cap B$$

NOTE 1. $P(A|B)$ reduces Ω to the part that B has occurred.

NOTE 2. $P(A|B) \neq P(B|A)$

NOTE 3. $P(A|B) = \frac{P(A \cap B)}{P(B)}$ is

a legitimate probability.

How to check?

• Axiom 1 : $P(A|B) = \frac{P(A \cap B)}{P(B)} \geq 0$

$$\because A \cap B \subseteq B$$

$$\therefore P(A \cap B) \leq P(B)$$

$$\therefore P(A|B) \leq 1.$$

• Axiom 2 : $P(\Omega | B) = \frac{P(\Omega \cap B)}{P(B)}$

$$= \frac{P(B)}{P(B)} = 1$$

- Axiom (3): if $A_1, A_2, \dots, A_n$ are disjoint, then

$$P\left(\bigcup_{j=1}^n A_j \mid B\right) \neq \sum_{j=1}^n P(A_j \mid B)$$

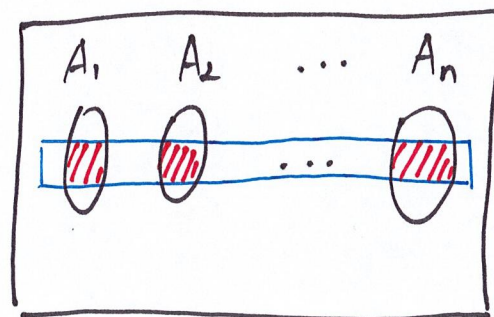
$$\text{LHS} = \frac{P\left(\bigcup_{j=1}^n A_j \cap B\right)}{P(B)}$$

$$= \frac{P\left(\bigcup_{j=1}^n (A_j \cap B)\right)}{P(B)}$$

$$= \frac{\sum_{j=1}^n P(A_j \cap B)}{P(B)}$$

$$= \sum_{j=1}^n \frac{P(A_j \cap B)}{P(B)} = \sum_{j=1}^n P(A_j \mid B) = \text{RHS}$$

distributive law



NOTE 4. $P(A|B) = 1 - P(A^c|B)$

NOTE 5. $P(A|B) \neq 1 - P(A|B^c)$

$$P(A|B) \neq 1 - P(A^c|B^c)$$

NOTE 6. If $A \perp B$, then

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot \cancel{P(B)}}{\cancel{P(B)}}$$

$$= P(A)$$

NOTE 7. $P(A|B) \neq P(B|A)$

$$\bullet P(A \cap B) = P(A|B) \cdot P(B)$$

$$\bullet P(A \cap B) = P(B|A) \cdot P(A)$$

$$P(A|B) = P(B|A) \cdot \frac{P(A)}{P(B)}$$